

UNIT - II

①

SET THEORY.

Basic concepts of set theory :- A set is any well-defined collection of objects called the elements (δ) members of the set. A collection of well-defined objects is called a set. Each of the objects in the set is called a member of an element of the set.

Ex: Books, cities, numbers, animals etc.

Elements of a set are usually denoted by lower-case letters while sets are denoted by capital letters.

Ex: Natural Numbers $N = \{1, 2, 3, 4, \dots\}$

The vowels $O = \{a, e, i, o, u\}$

Formalization of sets :- There are two ways by which the sets can be described.

1. Roster Notation :- Roster Notation is a complete list of all the elements of the set.

Ex: $B = \{2, 4, 6, 8, \dots, 40\}$

2. Set-Builder Notation :- It is used the roster method is impossible.

Ex: $B = \{x / 2 \leq x \leq 40, \text{ and 'x' is Even}\}$.

The vertical bar '/' is read as such that.

Symbol	Meaning
$\in$	Belongs to
$\forall$	for all
$\exists$	There Exist

Sub set :- If Every element in a set A is also element of a set B then A is called subset of B. It is denoted by $A \subseteq B$.

Note : The null set is a subset of Every set and Every set is a subset of itself.

i.e. $\phi \subseteq A$ and $A \subseteq A$.

Ex: If $A = \{1, 3, 4, 5, 8, 9\}$, $B = \{1, 2, 3, 5, 7\}$, $C = \{1, 5\}$
Then $C \subseteq A$ and $C \subseteq B$.

proper subset :- If A is a subset of B and 'B' is not subset of A then A is called a proper subset of B it is denoted by $A \subset B$.

If A is subset of B then 'B' is called a super set of A and it is denoted by $B \supset A$.

Note : Two different sets are Equal i.e. $A = B$ if and only if $A \subset B$ and $B \subset A$.

Types of set :-

1. Null set :- The set with no elements is called an Empty set (or) Null set. A null set is denoted by ϕ .
The null set is a subset of Every set i.e. $\phi \subset A$

2. countable set :- A countable set is a set with the same cardinality as some subset of the set of natural numbers.

Ex: $R = \{x/x \text{ is a natural number}\}$

$\Rightarrow$ 'R' is an infinite set

$A = \{x/x \text{ is a Even number between '2' and 40}\}$

$\Rightarrow$ A is a finite set.

3. uncountable set :- The uncountability of ⁽²⁾ a set is closely related to its cardinal number, a set is uncountable if its cardinal number is larger than that of the natural numbers.

4. Universal set :- In many discussions all the sets are considered to be subsets of one particular set. This set is called the Universal set for that discussion. The universal set is denoted by ' μ ' (or) ' U '

5. power set :- The set of all subsets of a set ' A ' is called the power set of A . and it is denoted by $P(A)$. If A has ' n ' elements in it then $P(A)$ has 2^n elements.

Ex: If $A = \{1, 2, 3\}$ then
 $P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

6. Equivalent sets :- Two sets are said to be equivalent if there is a one-one correspondence between the elements of the two sets.

Ex: If $A = \{a, b, c\}$, $B = \{\alpha, \beta, \gamma\}$ then ' A ' and ' B ' are Equivalent sets.

7. Disjoint sets :- Two sets are said to be disjoint if they have no element in common.

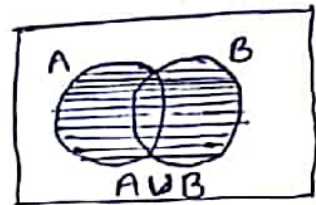
Ex: If $A = \{1, 2, 3\}$, $B = \{6, 7, 8, 9\}$ then ' A ' and ' B ' are disjoint sets. Here $A \cap B = \phi$.

Union of two sets :- The union of two sets ' A ' and ' B ' is the set whose elements are all of the elements "in A " or "in B " or "in both". The union of sets ' A ' and ' B '

is denoted by $A \cup B$ is read as "A Union 'B'".

EX: Let $A = \{1, 2, 3, 4\}$ $B = \{3, 4, 5, 6, 7\}$

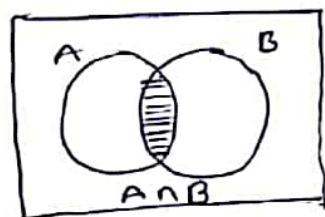
$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$



Intersection of two sets :- The intersection of two sets 'A' and 'B' is the set whose elements are all of the elements common to both 'A' and 'B'. The intersection of the sets of 'A' and 'B' is denoted by $A \cap B$ is read as "A intersection B".

EX: Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$

$$A \cap B = \{3, 4\}$$

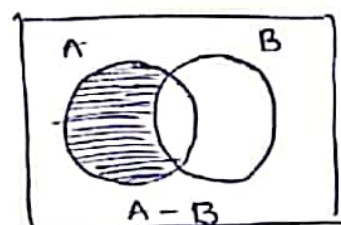


Difference of sets :- If A and B are subsets of the universal set μ then the Complement of B in A is the set of all elements in A which are not in B. and it is denoted by $A - B$

$$A - B = \{x / x \in A \text{ and } x \notin B\}$$

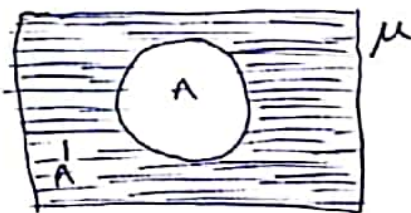
EX: If $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4\}$

$$A - B = \{1, \cancel{2}, \cancel{3}, \cancel{4}, 5\} = \{1, 5\}$$



Complement of a set :- If ' μ ' is a Universal set containing the set A then $\mu - A$ is called the complement of A and it is denoted by A' .

$$\mu - A = A' = \{x / x \notin A\}$$



EX: Let $\mu = \{1, 2, 3, 4, 5, \dots\}$, $A = \{1, 2, 3, 4\}$

$$A' = \mu - A = \{5, 6, 7, 8, \dots\}$$

$$B' = \mu - B = \{1, 3, 5, 7, 9, 10, \dots\}$$

Inclusion - Exclusion principle :- The inclusion-⁽³⁾

Exclusion principle is a counting technique which generalizes the familiar method of obtaining the number of elements in the Union of two finite sets. Symbolically expressed as

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

where $n(A)$ = number of elements in A

$n(B)$ = number of elements in B.

$n(A \cap B)$ = no. of elements both 'A' and 'B'

$n(A \cap B)$ = no. of elements in A and B with common.

For any finite sets A, B, C we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Ex: Find the no. of mathematics students at a college taking at least one of the languages Telugu, Hindi and English given the following data.

65 study Telugu, 20 study Telugu and Hindi
45 study Hindi, 25 study Telugu and English
42 study English, 15 study Hindi and English.
8 study all three languages.

Sol: Given $n(T) = 65$, $n(H) = 45$, $n(E) = 42$

$$n(T \cap H) = 20, \quad n(T \cap E) = 25, \quad n(H \cap E) = 15,$$

$$n(T \cap H \cap E) = 8$$

By inclusion - Exclusion principle

$$n(T \cup H \cup E) = n(T) + n(H) + n(E) - n(T \cap H) - n(T \cap E) - n(H \cap E) + n(T \cap H \cap E)$$

$$= 65 + 45 + 42 - 20 - 25 - 15 + 8$$

$$= 100 //$$

Cartesian product :- cartesian product is a direct product of sets. Let us consider two sets A and B the set of all order pair (a, b) where $a \in A$ and $b \in B$ is called the product of A and B and is denoted by $A \times B$.

Ex: Let $A = \{1, 2, 3\}$ $B = \{4, 5\}$ then

$$A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

$$B \times A = \{(4, 1), (5, 1), (4, 2), (5, 2), (4, 3), (5, 3)\}$$

$$\therefore A \times B \neq B \times A$$

Laws of set theory :-

1. Identity Laws: $A \cup \phi = A$ and $A \cap \mu = A$
2. Domination Laws: $A \cup \mu = \mu$ and $A \cap \phi = \phi$
3. Idempotent Laws: $A \cup A = A$ and $A \cap A = A$
4. Complementation Law: $(A')' = A$
5. Commutative Laws: $A \cup B = B \cup A$ and $A \cap B = B \cap A$
6. Associative Laws: $A \cup (B \cap C) = (A \cup B) \cap C$
 $A \cap (B \cup C) = (A \cap B) \cup C$
7. Distributive Laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
8. DeMorgan's Laws: $(A \cup B)' = A' \cap B'$
 $(A \cap B)' = A' \cup B'$
9. Absorption Laws: $A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$
10. Complement Laws: $A \cup A' = \mu$ and $A \cap A' = \phi$

Ex: prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

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Proof: Let 'x' is an arbitrary element of $A \cup (B \cap C)$

Then $x \in A \cup (B \cap C) \Leftrightarrow x \in A$ (or) $x \in (B \cap C)$

$\Leftrightarrow x \in A$ or $(x \in B \text{ and } x \in C)$

$\Leftrightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$

$\Leftrightarrow x \in (A \cup B) \text{ and } x \in (A \cup C)$

$\Leftrightarrow x \in (A \cup B) \cap (A \cup C)$

Hence $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Ex: prove that $(A \cap B)^c = A^c \cup B^c$

Proof: Let $x \in (A \cap B)^c$

$\Leftrightarrow x \notin A \cap B$

$\Leftrightarrow \sim (x \in A \cap B) \Leftrightarrow \sim (x \in A \text{ and } x \in B)$

$\Leftrightarrow \sim (x \in A) \text{ or } \sim (x \in B)$

$\Leftrightarrow x \notin A \text{ or } x \notin B \Leftrightarrow x \in A^c \text{ or } x \in B^c$

$\Leftrightarrow x \in A^c \cup B^c$

Hence $(A \cap B)^c = A^c \cup B^c$.

Ex: prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by using member-

ship table.

Proof:

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
Y	Y	Y	Y	Y	Y	Y	Y
Y	Y	N	Y	Y	Y	N	Y
Y	N	Y	Y	Y	N	Y	Y
Y	N	N	N	N	N	N	N
N	Y	Y	Y	N	N	N	N
N	Y	N	Y	N	N	N	N
N	N	Y	Y	N	N	N	N
N	N	N	N	N	N	N	N

Relations :- Any set of order pairs defines a binary relation. and it is arbitrary association of elements within a set or with elements of another set.

Mathematically let A and B are any two sets. A relation from A to B is a subset of $A \times B$.

Ex: $A = \{a, b\}$ and $B = \{a\}$

$$A \times B = \{(a, a), (b, a)\}$$

Any subset of $A \times B$ is called relation, i.e. $(a, a) \in A \times B$
 (a, a) is a relation.

Note: 1. $(a, b) \in R$ then we say "a is related to b" and it can be written as " $a R b$ " (or) $R(a, b)$
2. $(a, b) \notin R$ then we say "a is not related to b" and it can be written as " $a \not R b$ ".

ordered pair :- If $a \in A$ and $b \in B$ then the ordered pair is the set $\{\{a\}, \{a, b\}\}$ consisting of the pair $\{a, b\}$ and the singleton $\{a\}$. It is represented by (a, b) .

In the order pair (a, b) , the element 'a' is called the first element and 'b' is called the second element.

If (a, b) and (a', b') are two ordered pairs then.

$$(a, b) = (a', b') \Rightarrow a = a' \text{ and } b = b'$$

Note : 1. (a, b) and (b, a) are two different ordered pairs

$$(a, b) \neq (b, a)$$

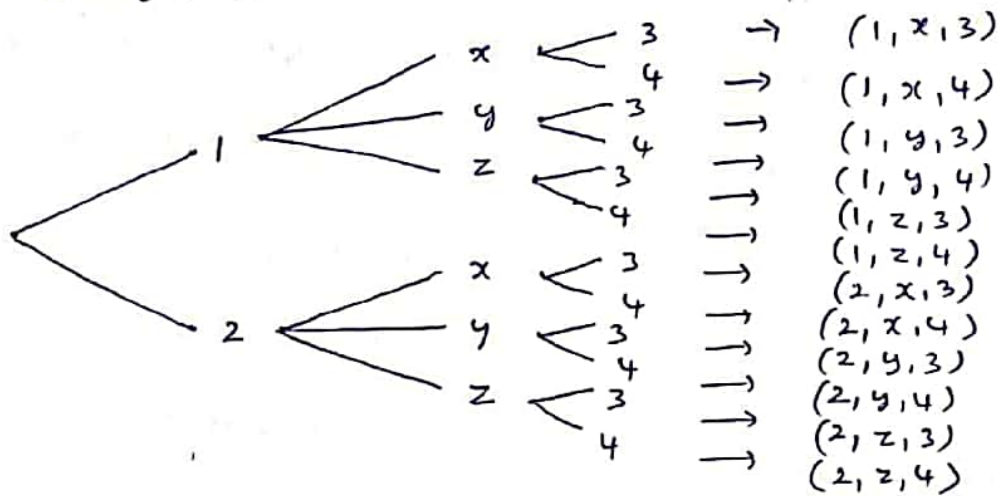
2. The two objects in an ordered pairs do not have to be distinct. i.e. (a, a) is a well-defined ordered pair.

Ex: Let $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$ then

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find $A \times B \times C$.

Sol: $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$



Domain and Range of relations :-

The domain of a relation R is the set of all first elements of the ordered pair which belongs to R and the Range of ' R ' is the set of second element.

1. $\text{Domain}(R) = \{a \in A / aRb \text{ for some } b \in B\}$

2. $\text{Range}(R) = \{b \in B / aRb \text{ for some } a \in A\}$.

Ex: If $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ ~~and~~ ^{then} find domain and Range. If $R = \{(1, a), (1, c), (2, b), (3, a)\}$.

Sol: Given $A = \{1, 2, 3\}$, $B = \{a, b, c\}$

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}$

Given Relation $R = \{(1, a), (1, c), (2, b), (3, a)\}$

Domain R = The set of first elements in R
 $= \{1, 2, 3\}$

Range R = The set of second elements in R
 $= \{a, c, b\}$.

properties of Binary Relations :-

1. Universal relation :- A relation R on a set A is Universal if $R = A \times A$.

Ex: If $A = \{1, 2, 3\}$ then $A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$ is the Universal relation on A .

2. void relation :- A relation R on a set A is void relation if R is the null set. ϕ .

Ex: If $A = \{3, 4, 5\}$ then $R = \{a + b > 10\}$ is a null set

3. Identity relation :- A relation R on a set A is Identity relation If $R = \{(a, a) / a \in A\}$ and it is denoted by I_A . Ex: If $A = \{1, 2, 3\}$ then $R = \{(1, 1), (2, 2), (3, 3)\}$.

4. Inverse relation :- If ' R ' is any relation from a set ' A ' to set ' B ' the inverse of R is denoted by R^{-1} .

$$R^{-1} = \{(y, x) / y \in B, x \in A, (x, y) \in R\}$$

Ex: If $A = \{2, 3, 5\}$, $B = \{6, 8, 10\}$ then $R = \{(2, 6), (2, 8), (2, 10), (3, 6), (5, 10)\}$

$$\text{Now } R^{-1} = \{(6, 2), (8, 2), (10, 2), (6, 3), (10, 5)\}$$

5. Reflexive relation :- A relation on a set A is reflexive if aRa for every $a \in A$ i.e. If $(a, a) \in R, \forall a \in A$

Ex: If $A = \{1, 2, 3, 4\}$ then $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

$\therefore$ ' R ' is called reflexive relation. otherwise it is called Irreflexive. Ex: $R = \{(1, 1), (2, 2), (2, 3), (4, 4)\}$.

6. Symmetric relation :- A relation R on a set A is called symmetric if $(b, a) \in R$ whenever $(a, b) \in R$ for all $a, b \in A$. (6)

7. Anti symmetric relation :- R is an anti symmetric if there exists (a, b) and $(b, a) \in R$ then $a = b$.

8. Transitive relation :- A relation R on a set A is transitive if whenever $a R b$ and $b R c$ then $a R c$.
i.e. $(a, b), (b, c) \in R$ then $(a, c) \in R$.

EX: If $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3)\}$ then find if relation is reflexive, symmetric and transitive.

Sol: $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3)\}$

i) Reflexive : $R \cup \{(2, 2), (4, 4)\}$

$\therefore R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3), \underline{(2, 2)}, \underline{(4, 4)}\}$

ii) Symmetric :- $R \cup \{(4, 2), (3, 4)\}$

$\therefore R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3), \underline{(4, 2)}, \underline{(3, 4)}\}$

iii) Transitive :- $R \cup \{(2, 3), (4, 1)\}$

$\therefore R = \{(1, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 3), \underline{(2, 3)}, \underline{(4, 1)}\}$

EX: If $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$ then draw its directed graph.

Sol: $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$



Matrix representation of relations :-

Let A and B are both finite sets and ' R ' is a relation from A to B . Then R may be represented as a matrix called the relation matrix of R . It is denoted by M_R .

Let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$ are two finite sets containing ' m ' and ' n ' elements respectively.

$M_R = [m_{ij}]_{m \times n}$ is defined by

$$m_{ij} = \begin{cases} 0 & \text{if } (a_i, b_j) \in R \\ 1 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

Ex: Let $A = \{1, 2, 3, 4\}$ and $R = \{(x, y) / x > y\}$ find M_R .

Sol: Given $A = \{1, 2, 3, 4\}$

$$R = \{(4, 1), (4, 3), (4, 2), (3, 1), (3, 2), (2, 1)\}$$

The corresponding relation matrix for the R are given

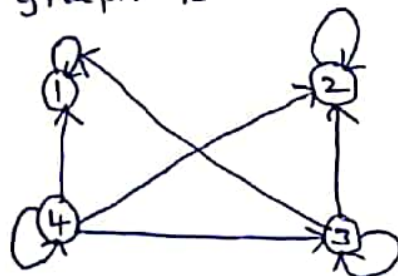
$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}_{4 \times 4}.$$

Ex: If $A = \{1, 2, 3, 4\}$ defined by $R = (1, 1), (2, 2), (3, 3), (4, 4), (4, 3), (4, 2), (4, 1), (3, 2), (3, 1)\}$. Find the matrix and directed graph of relation R .

Sol: The matrix of relation R is a 4×4 matrix as

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}.$$

directed graph is



Equivalence relations :- A relation R in a set $\textcircled{7}$

A is called an Equivalence relation if it is reflexive, symmetric and transitive.

Let a, b, c are arbitrary elements of some set A . Then " $a \sim b$ " (or) " $a \equiv b$ " denotes ' a ' is Equivalent to ' b '.

i, Reflexive : $a \sim a$

ii) Symmetric : If $a \sim b$ then $b \sim a$

iii) Transitive : If $a \sim b$ and $b \sim c$ then $a \sim c$.

Ex: Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,4), (4,1), (4,4), (2,2), (2,3), (3,2), (3,3)\}$. prove that ' R ' is an Equivalence.

Sol: Given $R = \{(1,1), (1,4), (4,1), (4,4), (2,2), (2,3), (3,2), (3,3)\}$

i) Reflexive :- For any $x \in A \Rightarrow (x, x) \in R$
 $\therefore xRx \Rightarrow 'R' \text{ is reflexive.}$ $\left\{ \begin{array}{l} (1,1) \quad (2,2) \quad (3,3) \\ (4,4) \end{array} \right\}$

ii) Symmetric :- For any $x, y \in A \Rightarrow$ If $(x, y) \in R$ then $(y, x) \in R$
 $\therefore xRy \Rightarrow yRx \Rightarrow 'R' \text{ is symmetric.}$ $\left\{ \begin{array}{l} \because (1,4) \rightarrow (4,1) \text{ and} \\ (2,3) \rightarrow (3,2) \end{array} \right.$

iii) Transitive :- For any $x, y, z \in A$
If (x, y) and $(y, z) \in R$ then $(x, z) \in R$

$\therefore xRy \text{ and } yRz \Rightarrow xRz$. ' R ' is transitive.

Hence the relation ' R ' is an Equivalence Relation.

$\left\{ \begin{array}{l} \because (1,1), (1,4) \rightarrow (1,4) \\ (2,3), (3,2) \rightarrow (2,2) \\ (3,2), (2,3) \rightarrow (3,3) \\ \text{etc...} \end{array} \right.$

Congruence Relation :- congruence relation is an example of equivalence relation. Let 'a' and 'b' are any two integers and 'r' be a fixed positive integer then 'a' and 'b' are said to be congruent modulo 'r' i.e. Mathematically $a \equiv b \pmod{r}$ if 'r' divides $(a-b)$.

Ex: $83 \equiv 13 \pmod{5}$

since $83 - 13 = 70$ is divisible by '5'

Ex: prove that the relation "congruence modulo m" over the set of positive integers is an equivalence relation

Sol: Let $x, y \in \mathbb{N}$, $x \equiv y \pmod{m}$ if and only if $x - y$ is divisible by 'm'.

i) Let $x, y, z \in \mathbb{N}$ then

$$x - x = 0 \cdot m \Rightarrow x \equiv x \pmod{m} \text{ for all } x \in \mathbb{N}$$

ii) Let $x \equiv y \pmod{m}$. Then $x - y$ is divisible by 'm'

$$\Rightarrow -(x - y) = y - x \text{ is also divisible by 'm'}$$

$$\text{i.e. } y \equiv x \pmod{m}$$

$\therefore$ The relation $\equiv$ is symmetric.

iii) Let $x \equiv y \pmod{m}$ and $y \equiv z \pmod{m}$

i.e. $(x - y)$ and $(y - z)$ are divisible by 'm'

Now $(x - y) + (y - z)$ is divisible by 'm'.

$$\text{i.e. } x - z \text{ is divisible by 'm'}$$

$$\Rightarrow x \equiv z \pmod{m} \quad \therefore \text{The relation } \equiv \text{ is transitive}$$

Hence ' $\equiv$ ' is equivalence relation.

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Ex: Let 'R' denote a relation on the set of order pairs of positive integers such that $(x, y) R (u, v)$ iff $xv = yu$. Show that R is an Equivalence relation.

Sol: Let x, y, u and v be positive integers

Given $(x, y) R (u, v) \Leftrightarrow xv = yu$

i) since $xy = yx$ is true for all positive integers

$$\Rightarrow (x, y) R (x, y)$$

$\therefore$ The relation R is reflexive.

ii) Let $(x, y) R (u, v) \Rightarrow xv = yu$

$$\Rightarrow yu = xv \Rightarrow uy = vx$$

$$\Rightarrow (u, v) R (x, y)$$

$\therefore$ The relation 'R' is symmetric.

iii) Let x, y, u, v, m and n are positive integers

Let $(x, y) R (u, v)$ and $(u, v) R (m, n)$

$$\Rightarrow xv = yu \quad \text{and} \quad un = vm$$

$$\Rightarrow xvn = yuvm \Rightarrow xn = ym \quad \therefore \text{by cancelling } uv$$

$$\Rightarrow (x, y) R (m, n)$$

$\therefore$ The relation 'R' is transitive.

Hence The relation 'R' is Equivalence relation.

Ex: N is the set of Natural numbers. The relation 'R' is defined by $(a, b) R (c, d) \Leftrightarrow a+d = b+c$. Prove that

R is an Equivalence relation.

Sol:

Let a, b, c and d be natural numbers.

Given $(a, b) R (c, d) \Leftrightarrow a+d = b+c$.

i) $(a, b) R (a, b) \Rightarrow a+b = b+a$

$\therefore$ R is reflexive.

$$\begin{aligned} \text{ii) } (a,b) R (c,d) &\Rightarrow a+d = b+c \\ &\Rightarrow b+c = a+d \\ &\Rightarrow c+b = d+a \Rightarrow (c,d) R (a,b) \end{aligned}$$

$\therefore 'R'$ is symmetric

$$\text{iii) } (a,b) R (c,d) \text{ and } (c,d) R (m,n)$$

$$\Rightarrow a+d = b+c \text{ and } c+n = d+m$$

$$\Rightarrow a+d+c+n = b+c+d+m$$

$$\Rightarrow a+n = b+m \Rightarrow (a,b) R (m,n)$$

$\therefore 'R'$ is transitive

Hence ' R ' is an Equivalence relation.

Compatibility relation :- A Relation ' R ' in X is said to be a compatibility relation if it is reflexive and symmetric. So all Equivalence relations are compatibility relations and it is sometimes denoted by " $\sim$ ".

Composition of relations :- Let ' R ' be a relation from x to y and ' S ' be a relation from y to z . Then a relation written as $R \circ S$ is called a composite relation of ' R ' and ' S ' where $R \circ S = \{ (x,z) / x \in x, z \in z \text{ there exist } y \in y \text{ with } (x,y) \in R \text{ and } (y,z) \in S \}$.

Ex: Let $R = \{ (1,2), (3,4), (2,2) \}$ and $S = \{ (4,2), (2,5), (3,1), (1,3) \}$. Find $R \circ S$ and $S \circ R$.

Sol: Given $R = \{ (1,2), (3,4), (2,2) \}$
 $S = \{ (4,2), (2,5), (3,1), (1,3) \}$

$$\therefore R \circ S = \{ (1,5), (3,2), (2,5) \}$$

$$S \circ R = \{ (4,2), (3,2), (1,4) \} \quad \text{clearly } R \circ S \neq S \circ R.$$

partial ordering :- A binary relation 'R' in P ⑨
is called a partial order relation (or) a partial ordering in P iff R is reflexive, antisymmetric and transitive. ie

1. aRa for all $a \in P$
2. aRb and $bRa \Rightarrow a=b$
3. aRb and $bRc \Rightarrow aRc$

Ex: show that the relation "greater than or equal to" is a partial ordering on the set of integers..

sol: Let 'Z' be the set of all integers and the relation $R = "\geq"$.

i) since $a \geq a$ for every integer a,
the relation ' $\geq$ ' is reflexive

ii) Let 'a' and 'b' be any two integers
Let aRb and $bRa \Rightarrow a \geq b$ and $b \geq a \Rightarrow a=b$
 $\therefore$ The relation " $\geq$ " is antisymmetric.

iii) Let a, b and 'c' be any three integers.
Let aRb and $bRc \Rightarrow a \geq b$ and $b \geq c \Rightarrow a \geq c$
 $\therefore$ The relation ' $\geq$ ' is transitive.

Hence the relation ' $\geq$ ' is partial ordering on the set of integers $\therefore (Z, \geq)$ is a poset.

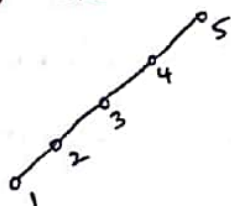
poset (partially ordered set) :- A set 'P' together with a partial ordering 'R' is called a partial ordered set or poset. The relation R is often denoted by the symbol ' $\leq$ ' then the ordered pair $(P, \leq)$ is called poset.

Hasse diagrams :- A partial order $\leq$ on a set P can be represented by means of a diagram known as Hasse diagram of $(P, \leq)$.

- i) Each element is represented by a small circle or dot
- ii) The circle for $x \in P$ is drawn below the circle for $y \in P$ if $x < y$ and a line is drawn between x and y if y covers 'x'.
- iii) If $x < y$ but y does not cover 'x' then x and y are not connected directly by a single line.

EX: Let $P = \{1, 2, 3, 4, 5\}$ and $\leq$ be the relation "Less than or Equal to" then the Hasse diagram is drawn below.

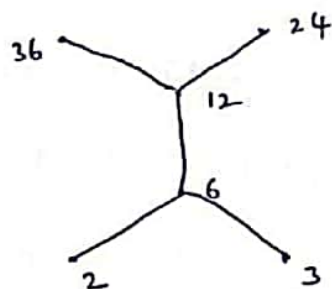
Sol:



It is totally order set.

EX: Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y . Draw the Hasse diagram of $(X, \leq)$.

Sol:



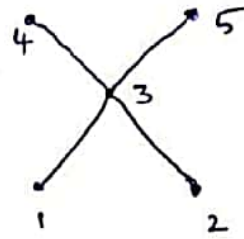
it is not a total order set.

EX: Draw the Hasse diagram for the relation R on $A = \{1, 2, 3, 4, 5\}$ whose relation matrix given below.

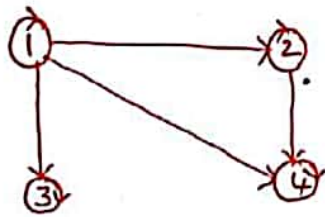
$$M_R = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Sol: $R = \{ (1,1), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,4), (5,5) \}$. (10)

Hasse diagram for M_R is

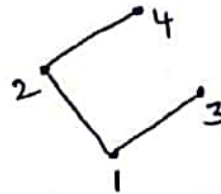


Ex: A partial order 'R' on the set $A = \{1, 2, 3, 4\}$ is represented by the following digraph. Draw the Hasse diagram for R.



Sol: By examining the given digraph we find that $R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4) \}$.

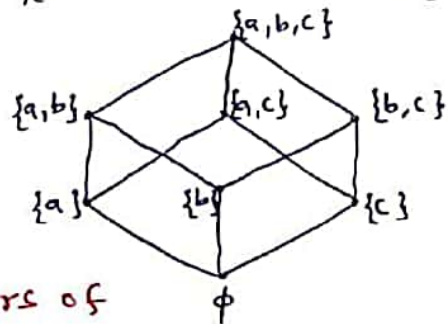
$\therefore$ Hasse diagram is



Ex: Draw the Hasse diagram for the partial ordering $\subseteq$ on the power set $P(S)$ where $S = \{a, b, c\}$

Sol: Given $S = \{a, b, c\}$
 $P(S) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \}$.

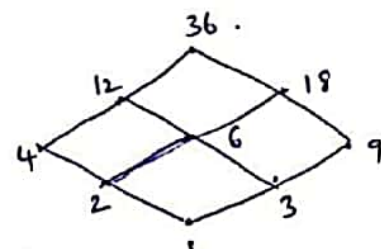
$\therefore$ Hasse diagram is



Ex: Draw the Hasse diagram representing the positive divisors of 36.

Sol: $D_{36} = \{ 1, 2, 3, 4, 6, 9, 12, 18, 36 \}$

$\therefore$ Hasse diagram is

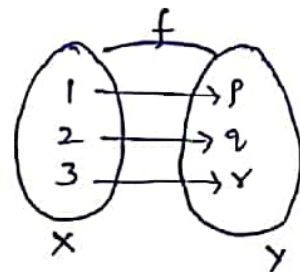


FUNCTIONS :- Let X and Y be any two sets. A relation ' f ' from ' X ' to ' Y ' is called a function if for every $x \in X$ there is a unique element $y \in Y$ such that $(x, y) \in f$.

Ex: Let $X = \{1, 2, 3\}$, $Y = \{p, q, r\}$ and $f = \{(1, p), (2, q), (3, r)\}$

Then $f(1) = p$, $f(2) = q$, $f(3) = r$

clearly ' f ' is a function from $X \rightarrow Y$



Domain and range of a function :-

If $f: X \rightarrow Y$ is a function then ' X ' is called the Domain of ' f ' and the set ' Y ' is called the Codomain of f . The range of ' f ' is defined as the set of all images under ' f '. It is denoted by $f(X) = \{y / \text{for some } x \text{ in } X, f(x) = y\}$

Ex: If the function ' f ' is defined by $f(x) = x^2 + 1$ on the set $\{-2, -1, 0, 1, 2\}$ find the range of ' f '.

Sol: Given $f(x) = x^2 + 1$

$$f(-2) = (-2)^2 + 1 = 5$$

$$f(-1) = (-1)^2 + 1 = 2$$

$$f(0) = 0 + 1 = 1 \quad f(2) = 4 + 1 = 5$$

$$f(1) = 1 + 1 = 2$$

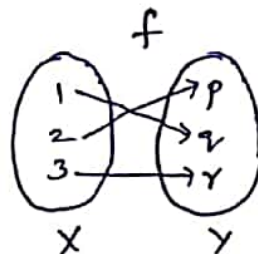
$\therefore$ The range of ' f ' = $\{1, 2, 5\}$.

Types of functions :-

1. one-to-one function (Injection) :- A mapping $f: X \rightarrow Y$ is called one-to-one if distinct elements of ' X ' are mapped into distinct elements of Y

i.e. f is one-to-one if

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \text{ for } x_1, x_2 \in X$$



Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x$, $\forall x \in \mathbb{R}$ is

(11)

one-one, since

$$f(x_1) = f(x_2) \Rightarrow 3x_1 = 3x_2 \Rightarrow x_1 = x_2, \forall x_1, x_2 \in \mathbb{R}.$$

2. onto function (Surjection) :- A mapping $f: X \rightarrow Y$

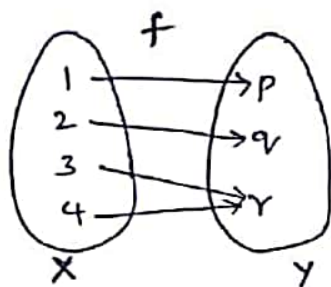
is called onto if the range set $R_f = Y$.

If $f: X \rightarrow Y$ is onto, then each element of Y is f -image of atleast one element of X .

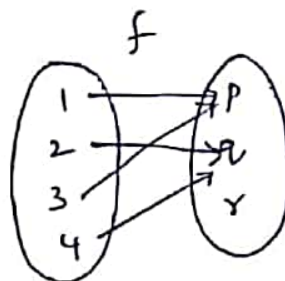
Mathematically $\forall y \in Y, \exists x \in X$ such that $f(x) = y$.

Note: If ' f ' is not onto then it is said to be into.

Ex:



Surjective



not surjective.

3. Bijection function :- A mapping $f: X \rightarrow Y$ is called

Bijection function if it is one-to-one and onto.

Such a mapping is also called a one-to-one correspondence between ' X ' and ' Y '.

Ex: show that a mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x+1$ for $x \in \mathbb{R}$ is a bijective map from $\mathbb{R}$ to $\mathbb{R}$.

Sol: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x+1$, for $x \in \mathbb{R}$.

i) one-to-one: Let ' x ' and ' y ' be any two elements

in $\mathbb{R}$ such that $f(x) = f(y)$

$$\Rightarrow 2x+1 = 2y+1$$

$$\Rightarrow 2x = 2y \Rightarrow x = y$$

$$\therefore f(x) = f(y) \Rightarrow x = y$$

$\therefore f$ is one-to-one.

ii) onto :- Let y' be any element in the domain R

$$\Rightarrow f(x) = y \Rightarrow 2x+1 = y \Rightarrow x = \frac{(y-1)}{2}$$

clearly $x = \frac{(y-1)}{2} \in R$. $\therefore f$ is onto

Hence ' f ' is a bijective map.

Composition of Functions :- Let $f: X \rightarrow Y$ and

$g: Y \rightarrow Z$ be two functions. Then the composition of ' f ' and ' g ' denoted by $g \circ f$ is the function from ' X ' to ' Z '

defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$.

Ex:- Let $f(x) = x+2$, $g(x) = x-2$, and $h(x) = 3x$ for $x \in R$ where R is the set of real numbers. Find $g \circ f$, $f \circ g$, $f \circ f$, $g \circ g$, $f \circ h$, $h \circ g$, $h \circ f$ and $f \circ g \circ h$.

Sol: Given $f(x) = x+2$, $g(x) = x-2$, $h(x) = 3x$

$$i) (g \circ f)(x) = g(f(x)) = g(x+2) = x+2-2 = x$$

$$ii) (f \circ g)(x) = f(g(x)) = f(x-2) = x-2+2 = x$$

$$iii) (f \circ f)(x) = f(f(x)) = f(x+2) = x+2+2 = x+4$$

$$iv) (g \circ g)(x) = g(g(x)) = g(x-2) = x-2-2 = x-4$$

$$v) (f \circ h)(x) = f(h(x)) = f(3x) = 3x+2 = 3x+2$$

$$vi) (h \circ g)(x) = h(g(x)) = h(x-2) = 3(x-2) = 3x-6$$

$$vii) (h \circ f)(x) = h(f(x)) = h(x+2) = 3(x+2) = 3x+6$$

$$viii) (f \circ g \circ h)(x) = f((g \circ h)(x)) = f(g(h(x))) = f(g(3x)) \\ = f(3x-2) = 3x-2+2 = 3x //$$

Ex: If $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ then (12)

find the composition of i) $f \circ f$ ii) $f \circ g$ iii) $g \circ f$.

Sol:

Inverse functions :- A function $f: X \rightarrow Y$ is said to be invertible if its inverse function f^{-1} is also function from the range of f into X .

If $f(x) = y$ then $x = f^{-1}(y)$.

Ex: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^3 - 2$. Find f^{-1} .

Sol: i) Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$
 $\Rightarrow x_1^3 - 2 = x_2^3 - 2 \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$

$\therefore f$ is one-to-one.

ii) If $f(x) = y$ then $y = x^3 - 2 \Rightarrow x^3 = y + 2$

$\Rightarrow x = \sqrt[3]{y+2} \therefore f$ is onto.

$\Rightarrow f^{-1}$ is invertible and $f^{-1}(x) = \sqrt[3]{x+2}$.

Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = ax + b$ for $a, b \in \mathbb{R}$. Show that f is invertible and find the inverse of f .

Sol: f is invertible and $f^{-1}(x) = \frac{x-b}{a}$.

Floor and ceiling functions :- Let x be a real number, then the least integer that is not less than x is called the ceiling of x . The ceiling of x is denoted by $\lceil x \rceil$.

Ex: $\lceil 2.15 \rceil = 3$, $\lceil \sqrt{5} \rceil = 3$, $\lceil -7.4 \rceil = -7$, $\lceil -2 \rceil = -2$

Let x be a real number, then the greatest integer that does not exceed x is called the 'Floor' of x . The floor of x is denoted by $\lfloor x \rfloor$.

Ex: $\lfloor 5.14 \rfloor = 5$ $\lfloor \sqrt{5} \rfloor = 2$, $\lfloor -7.6 \rfloor = -8$

Recursive function :- A recursive function is a function which is defined in terms of itself. We can also define functions recursively.

A recursive definition has two parts.

1. definition of the smallest argument i.e. $f(0)$ or $f(1)$
2. definition of $f(n)$, given $f(n-1)$, $f(n-2)$ etc...

Ex: An example of a recursively defined function.

$$f(0) = 3, \quad f(n+1) = 2f(n) + 3. \quad \text{Then}$$

$$f(1) = 2f(0) + 3 = 2 \cdot (3) + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 \cdot (9) + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 \cdot (21) + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 \cdot (45) + 3 = 93 \dots \dots$$

Ex: calculate the values of the following recursively defined function $f(0) = 1$ and $f(n) = n \cdot f(n-1)$.

Sol: Given $f(0) = 1$ and $f(n) = n \cdot f(n-1)$

$$f(1) = 1 \cdot f(0) = 1 \cdot 1 = 1$$

$$f(2) = 2 \cdot f(1) = 2 \cdot 1 = 2$$

$$f(3) = 3 \cdot f(2) = 3 \cdot 2 = 6$$

$$f(4) = 4 \cdot f(3) = 4 \cdot 6 = 24 \dots \dots$$

this is the recursive definition of the factorial function, $f(n) = n!$.

Ex: $f(1) = 1$, $f(2) = 1$ and $f(n) = f(n-2) + f(n-1)$

Ex: $f(0) = 0$, $f(n) = f(n-1) + 2n - 1$

Ex: $f(0) = 5$, $f(n) = f(n-1) + 2$.

Algebraic structures

(17)

Binary operation :- Let 'S' be a non empty set. If $f: S \times S \rightarrow S$ is a mapping, then 'f' is called a binary operation or binary composition in 'S'. The symbols $+$, $\cdot$, $*$, $\circ$, $\oplus$ etc are used to denote binary operations

Ex: For $a, b \in S \Rightarrow a + b \in S \Rightarrow '+'$ is a binary operation.

For $a, b \in S \Rightarrow a * b \in S \Rightarrow '*'$ is a binary operation.

This is said to be the closure property of the binary operation and the set 'S' is said to be closed with respect to the binary operation.

Algebraic structures :- A non empty set 'G' equipped with one or more binary operations is called an algebraic structure or an algebraic system. If '*' is a binary operation on G then the algebraic structure is written as $(G, *)$.

Ex: $(\mathbb{N}, +)$, $(\mathbb{Q}, -)$, $(\mathbb{R}, +)$ etc are algebraic structures

Semi group :- An algebraic structure $(G, *)$ is called a semi group if the binary operation '*' is a binary operation. Then $(G, *)$ is said to be a semi group if

- i, $a, b \in G \Rightarrow a * b \in G$ for all $a, b \in G$

ii, $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$

i, property is called closure property

ii, property is called Associative property.

Ex: The operation $*$ is defined by $a*b = a$ for all $a, b \in G$.
 Show that $(G, *)$ is a semi group.

Sol: Let $a, b \in G \Rightarrow a*b = a \in G$
 $\therefore$ ' $*$ ' is a binary operation in G

i) closure property :- Let $a, b \in G$ then $a*b = a \in G$

ii) Associative property :- Let $a, b, c \in G$

$$a*(b*c) = a*b = a$$

$$(a*b)*c = a*c = a$$

$$\therefore a*(b*c) = (a*b)*c$$

' $*$ ' is associative in G

Hence $(G, *)$ is a semi group.

Ex: The operation $*$ is defined by $a*b = a+b-ab$ for all $a, b \in \mathbb{Z}$ show that $(\mathbb{Z}, *)$ is a semi group.

Sol: Let $a, b \in \mathbb{Z} \Rightarrow a*b = a+b-ab \in \mathbb{Z}$

$\therefore$ ' $*$ ' is a binary operation in G

i) closure property :- Let $a, b \in G$ then $a*b = a+b-ab \in G$

ii) Associative property :- Let $a, b, c \in G$

$$a*(b*c) = a*(b+c-bc)$$

$$= a+(b+c-bc) - a(b+c-bc)$$

$$= a+b+c - \cancel{ab} - bc - ab - ac + abc$$

$$(a*b)*c = (a+b-ab)*c$$

$$= a+b-ab+c - (a+b-ab).c$$

$$= a+b+c - ab - ac - bc - abc$$

$$\therefore a*(b*c) = (a*b)*c$$

Hence $(\mathbb{Z}, *)$ is a semi group.

Ex: 'Q' is the set of rational numbers, * is a binary operation defined on Q such that $a * b = a - b + ab$ for $a, b \in Q$. Then $(Q, *)$ is not a semi group.

Sol: Let $a, b \in Q \Rightarrow a * b = a - b + ab \in Q$

'*' is a binary operation in Q.

i) closure property: Let $a, b \in Q \Rightarrow a * b = a - b + ab \in Q$

ii) Associative property: Let $a, b, c \in Q$

$$\begin{aligned} a * (b * c) &= a * (b - c + bc) \\ &= a - (b - c + bc) + a(b - c + bc) \\ &= a - b + c - bc + ab - ac + abc \end{aligned}$$

$$\begin{aligned} (a * b) * c &= (a - b + ab) * c \\ &= (a - b + ab) - c + (a - b + ab) \cdot c \\ &= a - b - c + ab + ac - bc + abc \end{aligned}$$

$$\therefore a * (b * c) \neq (a * b) * c$$

Hence $(Q, *)$ is not a semi group.

Ex: Let $(A, *)$ be a semi group. show that for a, b, c in A if $a * c = c * a$ and $b * c = c * b$ then $(a * b) * c = c * (a * b)$.

Sol: Given $(A, *)$ is semi group
 $a * c = c * a$ and $b * c = c * b$ for $a, b, c \in A$

$$\begin{aligned} \text{Now } (a * b) * c &= a * (b * c) &< \because \text{Associative} \\ &= a * (c * b) &< \because c * b = b * c \\ &= (a * c) * b &< \because \text{Associative} \\ &= (c * a) * b &< \because a * c = c * a \\ &= c * (a * b) &< \because \text{Associative} \end{aligned}$$

$$\therefore a * (b * c) = c * (a * b)$$

Ex: Show that the binary operation $*$ defined on $(R, *)$ where $x * y = \max(x, y)$ is a semi group.

Sol: Let $x, y \in R \Rightarrow x * y = \max(x, y) \in R$

' $*$ ' is a binary operation in R

i) closure property : Let $x, y \in R \Rightarrow x * y = \max(x, y) \in R$

ii) Associative property :- Let $x, y, z \in R$

$$\begin{aligned}(x * y) * z &= \max(x, y) * z \\ &= \max(\max(x, y), z) = \max(x, y, z)\end{aligned}$$

$$\begin{aligned}x * (y * z) &= x * \max(y, z) \\ &= \max(x, \max(y, z)) = \max(x, y, z)\end{aligned}$$

$$\therefore x * (y * z) = (x * y) * z$$

Hence $(R, *)$ is a semi group.

Ex:- show that the binary operation $*$ defined on $(R, *)$ where $x * y = x^y$ is not a semi group.

Sol: Let $x, y \in R \Rightarrow x * y = x^y \in R$

' $*$ ' is a binary operation in R

i) closure property :- Let $x, y \in R \Rightarrow x * y = x^y \in R$

ii) Associative property : Let $a, b, c \in R$

$$\begin{aligned}(x * y) * z &= x^y * z \\ &= (x^y)^z = x^{yz}\end{aligned}$$

$$\begin{aligned}x * (y * z) &= x * y^z \\ &= x^{y^z}\end{aligned}$$

$$\therefore (x * y) * z \neq x * (y * z)$$

Hence $(R, *)$ is not a semi group.

Homomorphism of semi-group :-

Let $(S, *)$ and $(T, \circ)$ be any two semi-groups. A mapping $f: S \rightarrow T$ such that for any two elements $a, b \in S$ $f(a * b) = f(a) \circ f(b)$ is called a semi group homomorphism.

A homomorphism of a semi group into itself is called a semi group endomorphism.

Identity element :- Let G be a nonempty set and $*$ be a binary operation on G . If there exists an element $e \in G$ such that $a * e = e * a = a$ for $a \in G$. Then 'e' is called an identity element on G .

Monoid :- A semi group $(G, *)$ with an identity element with respect to the binary operation $*$ is known as a Monoid.

EX: $(\mathbb{Z}, +)$ is a Monoid and the identity is '0'.
 $(\mathbb{Z}, \cdot)$ is a Monoid and the identity is '1'.

Inverse element :- Let $(G, *)$ be an algebraic structure with the identity element 'e' in G w.r.t $*$. An element $a \in G$ is said to be invertible if there exist an element $x \in G$ such that $a * x = x * a = e$.

Group :- If G is a nonempty set and $*$ is a binary operation defined on G such that the following Laws are satisfied then $(G, *)$ is a group.

- i) closure Law : Let $a, b \in G \Rightarrow a * b \in G$
- ii, Associative Law : For $a, b, c \in G \Rightarrow a * (b * c) = (a * b) * c$
- iii) Identity Law : For $a \in G$ there exist $e \in G$ such that $a * e = e * a = a$.

iv) Inverse Law :- For $a \in G$ there exist an element $b \in G$ such that $a * b = b * a = e$, b is called an inverse of 'a'

Ex:- prove that $\{1, \omega, \omega^2\}$ is a group with respect to multiplication where $1, \omega, \omega^2$ are cube roots of unity.

Sol: we construct the composition table as follows.

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	$\omega^3 = 1$	ω

$$\omega^3 = 1, \omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$$

i) closure Law :- From the composition table it is clear that $(G, *)$ is closed w.r.to multiplication.

ii) Associative Law :- From the composition table it is clear that $(G, \cdot)$ is Associative w.r.to multiplication.

iii) Identity Law :- '1' is identity element in G such that $a \cdot 1 = 1 \cdot a = a, \forall a \in G$.

iv) Inverse Law :- Each element of G is invertible
 $1 \cdot 1 = 1 \cdot 1 = 1 \Rightarrow '1'$ is its own inverse
 $\omega \cdot \omega^2 = \omega^3 = 1 \Rightarrow \omega^2$ is the inverse of ω .

Hence $(G, \cdot)$ is a group.

Note :- $(G, \cdot)$ is a group and $a \cdot b = b \cdot a, \forall a, b \in G$ that is commutative Law holds in G with respect to the binary operation. then $(G, \cdot)$ is an abelian group.

Ex:- show that the set $G = \{1, -1, i, -i\}$ where $i^2 = -1$ is an abelian group with respect multiplication as a binary operation.

Sol: we construct the composition table as follow

i) closure Law :- From the table
 $(G, \cdot)$ is closed w.r. to multiplication

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

ii) Associative Law :- For any three elements $a, b, c \in G \Rightarrow$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

$$\text{since } 1 \cdot (-1 \cdot i) = 1 \cdot -i = -i$$

$$(1 \cdot -1) \cdot i = -1 \cdot i = -i$$

$\therefore$ Associative Law is satisfied.

iii) Identity Law :- '1' is the identity element in $(G, \cdot)$
 such that $a \cdot 1 = 1 \cdot a = a, \forall a \in G$.

iv) Inverse Law :- $1 \cdot 1 = 1 \cdot 1 = 1 \Rightarrow$ '1' is inverse of '1'

$$(-1) \cdot (-1) = (-1) \cdot (-1) = 1 \Rightarrow -1 \text{ is the inverse of } -1$$

$$i \cdot -i = -i \cdot i = 1 \Rightarrow i \text{ is the inverse of } -i$$

Hence Inverse of every element in G exists.

v) Commutative Law :- For $a, b \in G \Rightarrow a \cdot b = b \cdot a$
 $1 \cdot 1 = 1 \cdot 1 = 1, -1 \cdot 1 = 1 \cdot -1 = -1$
 $i \cdot 1 = 1 \cdot i = i, -i \cdot i = i \cdot (-i) = 1$ etc...

Commutative Law is satisfied.

Hence $(G, \cdot)$ is an abelian group.

Ex: Prove that the set 'Z' of all integers with binary operation '*' defined by $a * b = a + b + 1, \forall a, b \in Z$ is an abelian group.

Sol: Given $a * b = a + b + 1, \forall a, b \in Z$

i) closure Law :- Let $a, b \in Z \Rightarrow a * b = a + b + 1 \in Z$
 $\therefore$ 'Z' is closed under '*'

ii) Associative Law :- Let $a, b, c \in Z$

$$(a * b) * c = (a + b + 1) * c$$

$$= a + b + 1 + c + 1 = a + b + c + 2$$

$$a * (b * c) = a * (b + c + 1)$$

$$= a + b + c + 1 + 1 = a + b + c + 2$$

$$\therefore a * (b * c) = (a * b) * c.$$

iii) Identity Law :- Let $a \in \mathbb{Z}$, there exist $e \in \mathbb{Z}$

such that $a * e = e * a = a$

$$a + e + 1 = a \Rightarrow e + 1 = 0 \Rightarrow e = -1$$

$\therefore e = -1$ is the identity element in $\mathbb{Z}$.

iv) Inverse Law :- Let $a \in \mathbb{Z}$, there exist $b \in \mathbb{Z}$

such that $a * b = e$

$$a + b + 1 = -1 \Rightarrow a + b = -2 \Rightarrow b = -2 - a$$

$\therefore b = -2 - a$ is an Inverse element in $\mathbb{Z}$.

v) Commutative Law :- For $a, b \in \mathbb{Z} \Rightarrow a * b = b * a$

$$a * b = a + b + 1 = b + a + 1 = b * a$$

$\therefore$ Commutative Law is satisfied.

Hence $(\mathbb{Z}, *)$ is an abelian group.

Ex: Show that the set $\mathbb{Q}_+$ of all positive rational numbers forms an abelian group under the composition defined by $*$ such that $a * b = \frac{ab}{3}$ for $a, b \in \mathbb{Q}_+$

sol: Let $a, b \in \mathbb{Q}_+ \Rightarrow a * b = \frac{ab}{3}$.

i) closure Law :- For $a, b \in \mathbb{Q}_+ \Rightarrow a * b = \frac{ab}{3} \in \mathbb{Q}_+$

ii) Associative Law : For $a, b, c \in \mathbb{Q}_+$

$$a * (b * c) = a * \frac{bc}{3} = \frac{a \cdot \frac{bc}{3}}{3} = \frac{abc}{9}$$

$$(a * b) * c = \frac{ab}{3} * c = \frac{\frac{ab}{3} \cdot c}{3} = \frac{abc}{9}$$

$\therefore a * (b * c) = (a * b) * c$, Associative Law satisfied.

iii) Identity Law :- Let $a \in Q_+$ there exist $e \in Q_+$ such that $a \cdot e = e \cdot a = a$.

$$\frac{ae}{3} = a \Rightarrow ae - 3a = 0 \Rightarrow a(e-3) = 0 \quad (\because a \neq 0)$$

$$\therefore e-3 = 0 \Rightarrow \boxed{e=3}$$

$\therefore e=3$ is the identity element in Q_+

iv) Inverse Law :- Let $a \in Q_+$ there exist $b \in Q_+$ such

that $a * b = e = b * a$

$$\frac{ab}{3} = e \Rightarrow \frac{ab}{3} = 3 \Rightarrow ab = 9 \Rightarrow \boxed{b = \frac{9}{a}}$$

$\therefore b = \frac{9}{a}$ is an inverse element in Q_+ .

v) Commutative Law :- For $a, b \in Q_+ \Rightarrow a * b = b * a$

$$a * b = \frac{ab}{3} = \frac{ba}{3} = b * a.$$

$\therefore$ Commutative Law is satisfied.

Hence $(Q_+, *)$ is an abelian group.

Ex: Prove that the set Q of rational numbers other than '1' with operation $\oplus$ such that $a \oplus b = a + b - ab$ for $a, b \in Q$ is abelian group. : practice problem

Ex: Consider the algebraic system $(G, *)$ where 'G' is the set of all non-zero real numbers and '*' is a binary operation defined by $a * b = \frac{ab}{4} \quad \forall a, b \in G$. Show that $(G, *)$ is an abelian group. : practice problem

Addition modulo 'm' :- If 'a' and 'b' are any two integers and 'r' is the least non-negative remainder obtaining by dividing the ordinary sum of 'a' and 'b' by 'm' then the addition modulo 'm' of 'a' and 'b' is 'r' symbolically $a +_m b = r, \quad 0 \leq r < m$

Ex: $20 +_6 5 = 1$ and $-15 +_6 3 = 3$

Multiplication modulo p: If 'a' and 'b' are any two integers and 'r' is the least non-negative remainder obtained by dividing the ordinary product of 'a' and 'b' by 'p' then the multiplication modulo 'p' of 'a' and 'b' is 'r' symbolically $a \times_p b = r$, $0 \leq r < p$

Ex: $5 \times_3 4 = 2$,

Ex: show that the set $G = \{0, 1, 2, 3, 4\}$ is an abelian group with respect to addition modulo '5'.

sol: we form the composition table as follows.

i) closure law :- since all the entries in the composition table are elements in G.

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\therefore$ closure Law satisfied.

ii) Associative Law :- For $a, b, c \in G$

$$(a +_5 b) +_5 c = a +_5 (b +_5 c)$$

$$(1 +_5 3) +_5 4 = 1 +_5 (3 +_5 4) = 3 \text{ etc.}$$

iii) Identity Law :- clearly $0 \in G$ is the identity element
 $0 +_5 a = a \equiv a +_5 0$, $\forall a \in G$

iv) Inverse Law :- From the table

'0' is its own inverse ; '1' is the inverse of '4'
 '2' is the inverse of 3 ; '3' is the inverse of '2'
 '4' is the inverse of '1'.

v) commutative Law :- From the table

$$a +_5 b = b +_5 a, \forall a, b \in G$$

Hence $(G, +_5)$ is an abelian group.

Ex: show that the set $G = \{1, 2, 3, 4\}$ is an abelian group with respect to multiplication modulo 5. (22)

Sol: The composition table for multiplication modulo 5 is

i) closure Law :- since all the entries in the composition table are elements in G .

$\times_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

$\therefore$ closure Law is satisfied.

ii) Associative Law :- For $a, b, c \in G$

$$(a \times_5 b) \times_5 c = a \times_5 (b \times_5 c)$$

$$(1 \times_5 3) \times_5 4 = 1 \times_5 (3 \times_5 4) = 2 \text{ etc.}$$

iii) Identity Law :- clearly $1 \in G$ is the identity element

$$1 \times_5 a = a = a \times_5 1 \Rightarrow \forall a \in G$$

iv) Inverse Law :- From the table

'1' is the inverse of '1'; '2' is the inverse of '3'
'3' is the inverse of '2'; '4' is the inverse of '4'

v) Commutative Law :- From the table

$$a \times_5 b = b \times_5 a, \forall a, b \in G$$

Hence $(G, \times_5)$ is an abelian group.

Ex: show that $G = \{x / x = 2^a 3^b \text{ for } a, b \in \mathbb{Z}\}$ is a group under multiplication.

Sol: Let $x, y, z \in G$, we can take

$$x = 2^p 3^q, \quad y = 2^r 3^s, \quad z = 2^l 3^m, \quad p, q, r, s, l, m \in \mathbb{Z}$$

i) closure Law :- For $x, y \in G \Rightarrow x \cdot y \in G$

$$\begin{aligned} \text{since } x \cdot y &= (2^p \cdot 3^q) \cdot (2^r \cdot 3^s) \\ &= 2^{p+r} \cdot 3^{q+s} \in G. \end{aligned}$$

ii) Associative Law :- For $x, y, z \in G$

$$\begin{aligned}
 x \cdot (y \cdot z) &= x \cdot (y^r z^s) \\
 &= 2^p 3^q (2^r \cdot 3^s \cdot 2^l 3^m) \\
 &= 2^p 3^q (2^{r+l} 3^{s+m}) \\
 &= 2^{p+r+l} \cdot 3^{q+s+m} \\
 &= (2^{p+r} \cdot 3^{q+s}) 2^l 3^m \\
 &= (2^p \cdot 3^q \cdot 2^r \cdot 3^s) 2^l 3^m = (x \cdot y) \cdot z
 \end{aligned}$$

$\therefore$ Associative is satisfied.

iii) Identity Law : Let $x \in G$

We know that $e = 2^0 3^0 \in G$, since $0 \in \mathbb{Z}$

$$\therefore x \cdot e = 2^p 3^q \cdot 2^0 3^0 = 2^{p+0} 3^{q+0} = 2^p 3^q = x$$

$$\therefore x \cdot e = e \cdot x = x$$

iv) Inverse Law :- Let $x \in G$

Now $y = 2^{-p} 3^{-q} \in G$, since $-p, -q \in \mathbb{Z}$

$$\therefore x \cdot y = 2^p 3^q \cdot 2^{-p} 3^{-q} = 2^{p-p} 3^{q-q} = 2^0 3^0 = e$$

$$\therefore x \cdot y = y \cdot x = e$$

Hence $(G, \cdot)$ is a group.

Ex: If $(G, *)$ is a group then $(a * b)^{-1} = b^{-1} * a^{-1}$, $\forall a, b \in G$

Sol: Let $a, b \in G$ and 'e' is the identity element in G

Let $a \in G \Rightarrow a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

$b \in G \Rightarrow b^{-1} \in G$ such that $b * b^{-1} = b^{-1} * b = e$

Now $a, b \in G \Rightarrow a * b \in G$ and $(a * b)^{-1} \in G$

Consider

$$\begin{aligned}
 (a * b) * (b^{-1} * a^{-1}) &= a * [b * (b^{-1} * a^{-1})] \\
 &= a * [(b * b^{-1}) * a^{-1}]
 \end{aligned}$$

$$= a * (e * a^{-1})$$

$$= a * a^{-1} = e$$

$$(b^{-1} * a^{-1}) * (a * b) = b^{-1} * [a^{-1} * (a * b)]$$

$$= b^{-1} * [(a^{-1} * a) * b]$$

$$= b^{-1} * (e * b)$$

$$= b^{-1} * b = e$$

$$\therefore (a * b) * (b^{-1} * a^{-1}) = (b^{-1} * a^{-1}) * (a * b) = e$$

$$\text{Hence } (a * b)^{-1} = b^{-1} * a^{-1}$$

Ex: Cancellation Laws holds in G

For all $a, b, c \in G$ then $a * b = a * c \Rightarrow b = c$ (L.C.L)
 $b * a = c * a \Rightarrow b = c$ (R.C.L)

Sol: G is a group

Let 'e' be the identity element in G .

$a \in G \Rightarrow a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

Consider $a * b = a * c$

$$\Rightarrow a^{-1} * (a * b) = a^{-1} * (a * c)$$

$$\Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c$$

$$\Rightarrow e * b = e * c \Rightarrow b = c$$

Now $b * a = c * a$

$$\Rightarrow (b * a) * a^{-1} = (c * a) * a^{-1}$$

$$\Rightarrow b * (a * a^{-1}) = c * (a * a^{-1})$$

$$\Rightarrow b * e = c * e \Rightarrow b = c //$$

Ex: If every element of a group G is its own inverse
 show that G is an abelian group.

Sol: Let $a, b \in G$ By Hypothesis $a^{-1} = a, b^{-1} = b$

$$\text{Then } ab \in G \Rightarrow (ab)^{-1} = ab$$

$$\text{Now } (ab)^{-1} = ab \Rightarrow b^{-1} a^{-1} = ab \Rightarrow ba = ab$$

$\therefore 'G'$ is an abelian group.

Ex: Prove that if $a^2 = a$ then $a = e$, 'a' being an element of a group G.

Sol: Let 'a' be an element of a group G such that $a^2 = a$

$$\text{Now } a^2 = a \Rightarrow aa = a$$

$$\Rightarrow (aa)a^{-1} = aa^{-1}$$

$$\Rightarrow a(aa^{-1}) = aa^{-1} \quad \left\{ \because aa^{-1} = e \right.$$

$$\Rightarrow ae = e \Rightarrow \boxed{a = e}$$

Ex: Show that in a group G, for $a, b \in G$, $(ab)^2 = a^2b^2 \Leftrightarrow G$ is an abelian.

Sol: Let $a, b \in G$ and $(ab)^2 = a^2b^2$ Then

$$(ab)^2 = a^2b^2 \Rightarrow (ab)(ab) = a^2b^2 = (aa)(bb)$$

$$\Rightarrow a(ba)b = a(ab)b$$

$$\Rightarrow ba = ab \quad (\text{by cancellation Law})$$

$$\Rightarrow 'G' \text{ is abelian}$$

Conversely, Let G is an abelian Then

$$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = (aa)(bb) = a^2b^2 //$$

Ex: If a, b are any two elements of a group $(G, \cdot)$ which commute. Show that 1. a^{-1} and 'b' commute
2. b^{-1} and 'a' commute 3. a^{-1} and b^{-1} commute

Sol: $(G, \cdot)$ is a group such that $ab = ba$

$$1. \quad ab = ba \Rightarrow a^{-1}(ab) = a^{-1}(ba)$$

$$\Rightarrow (a^{-1}a)b = a^{-1}(ba)$$

$$\Rightarrow eb = (a^{-1}b)a \Rightarrow b = (a^{-1}b)a$$

$$\Rightarrow ba^{-1} = (a^{-1}b)aa^{-1}$$

$$\Rightarrow ba^{-1} = (a^{-1}b)e$$

$$\Rightarrow ba^{-1} = a^{-1}b \Rightarrow 'a^{-1}' \text{ and 'b' commute.}$$

$$\begin{aligned}
 2. \quad ab = ba &\Rightarrow (ab)b^{-1} = (ba)b^{-1} & (24) \\
 &\Rightarrow a(bb^{-1}) = (ba)b^{-1} \\
 &\Rightarrow ae = b(ab^{-1}) \\
 &\Rightarrow a = b(ab^{-1}) \\
 &\Rightarrow b^{-1}a = b^{-1}(b(ab^{-1})) \\
 &\Rightarrow b^{-1}a = b^{-1}b(ab^{-1}) \Rightarrow b^{-1}a = e(ab^{-1}) \\
 &\Rightarrow b^{-1}a = ab^{-1}
 \end{aligned}$$

$\therefore b^{-1}$ and 'a' commute

$$\begin{aligned}
 3. \quad ab = ba &\Rightarrow (ab)^{-1} = (ba)^{-1} \Rightarrow b^{-1}a^{-1} = a^{-1}b^{-1} \\
 \therefore 'a^{-1}' \text{ and } 'b^{-1}' &\text{ are commute.}
 \end{aligned}$$

Order of an element :- Let $(G, *)$ be a group and $a \in G$ then the least positive integer 'n' if it exist such that $a^n = e$ is called the order of $a \in G$. The order of an element $a \in G$ is denoted by $O(a)$.

Ex: If $G = \{1, -1, i, -i\}$

$$1^1 = 1^2 = 1^3 = \dots = 1 \Rightarrow O(1) = 1$$

$$(-1)^2 = (-1)^4 = (-1)^6 = \dots = 1 \Rightarrow O(-1) = 2$$

$$i^4 = i^8 = \dots = 1 \Rightarrow O(i) = 4$$

$$(-i)^4 = (-i)^8 = \dots = 1 \Rightarrow O(-i) = 4.$$

Sub groups :- Let $(G, *)$ be a group and 'H' be a non empty subset of G . If $(H, *)$ is itself is a group then $(H, *)$ is called sub-group of $(G, *)$

Ex: $(\mathbb{Z}, +)$ is a sub group of $(\mathbb{Q}, +)$

$(\mathbb{N}, +)$ is not a sub group of the group $(\mathbb{Z}, +)$;

Ex: If $(G, *)$ is a group and $H \subseteq G$ then $(H, *)$ is a sub group of $(G, *)$ if and only if

- i) $a, b \in H \Rightarrow a * b \in H$
- ii) $a \in H \Rightarrow a^{-1} \in H$.

Sol: Let $(H, *)$ be a sub group of $(G, *)$

since $(H, *)$ is a group

By closure property we have $a, b \in H \Rightarrow ab \in H$

By inverse property $a \in H \Rightarrow a^{-1} \in H$.

Conversely Let $a, b \in H \Rightarrow ab \in H$ and $a \in H \Rightarrow a^{-1} \in H$.

since H is non empty

Let $a \in H \Rightarrow a^{-1} \in H$

$\therefore a \in H, a^{-1} \in H \Rightarrow aa^{-1} \in H \Rightarrow e \in H$, e is identity in G

$\Rightarrow 'e'$ is the identity in H .

' $*$ ' is associative in H follows from the fact that ' $*$ ' is associative in G .

Hence H itself is a group, $\therefore H$ is a sub group of ' G '.

Ex: If H_1 and H_2 are two subgroups of a group G . then

$H_1 \cap H_2$ is also a sub group of G .

Sol: Let H_1 and H_2 be two subgroups of a group G .

Let ' e ' be the identity element in G :

$\therefore e \in H_1$ and $e \in H_2 \Rightarrow e \in H_1 \cap H_2 \Rightarrow H_1 \cap H_2 \neq \emptyset$

Let $a \in H_1 \cap H_2$, and $b \in H_1 \cap H_2$

$a \in H_1, a \in H_2$ and $b \in H_1, b \in H_2$

Since H_1 is a sub group, $a \in H_1$ and $b \in H_1 \Rightarrow ab^{-1} \in H_1$

Similarly $ab^{-1} \in H_2 \therefore ab^{-1} \in H_1 \cap H_2$

we have $a \in H_1 \cap H_2, b \in H_1 \cap H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ is a sub group of G .